Mobile Application Development

Practical No. 11

Develop a program to implement Grid View, Image View, Scroll View, List View for any management system like library management/hotel management.

1 – Using Grid View, Image View, Scroll View, List View:-

- **activity_main.xml :-**

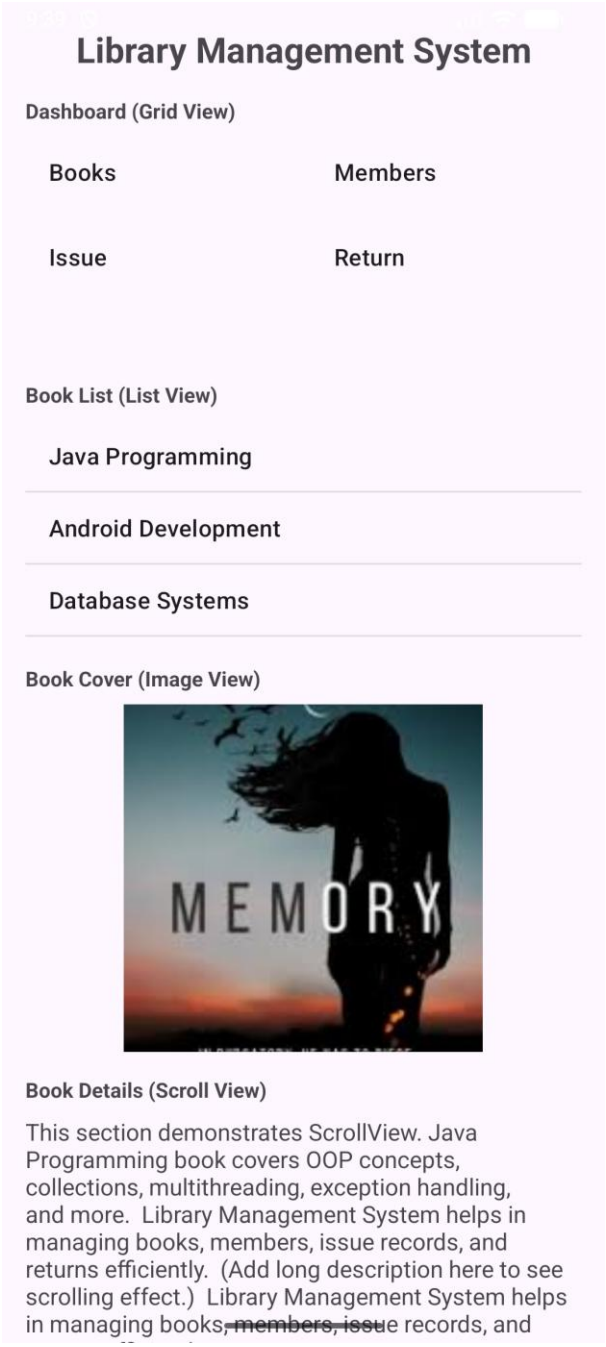
```
<?xml version="1.0" encoding="utf-8"?>

<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">


    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="16dp">


        <!-- ===== TITLE ===== -->

        <TextView
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Library Management System"
            android:textSize="24sp"
            android:textStyle="bold"
            android:gravity="center"
            android:paddingBottom="16dp"/>


        <!-- ===== GRID VIEW ===== -->

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Dashboard (Grid View)"
            android:textStyle="bold"
            android:paddingBottom="8dp"/>


        <GridView
            android:id="@+id/gridView"
            android:layout_width="match_parent"
            android:layout_height="150dp"
            android:numColumns="2"
```

```
        android:verticalSpacing="10dp"
        android:horizontalSpacing="10dp"
        android:gravity="center"/>

    <!-- ===== LIST VIEW ===== -->
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Book List (List View)"
        android:textStyle="bold"
        android:paddingTop="16dp"
        android:paddingBottom="8dp"/>

    <ListView
        android:id="@+id/listView"
        android:layout_width="match_parent"
        android:layout_height="150dp"/>

    <!-- ===== IMAGE VIEW ===== -->

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Book Cover (Image View)"
        android:textStyle="bold"
        android:paddingTop="16dp"
        android:paddingBottom="8dp"/>

    <ImageView
        android:id="@+id/imageView"
        android:layout_width="245dp"
        android:layout_height="237dp"
        android:layout_gravity="center"
        android:scaleType="centerCrop"
        android:src="@android:drawable/ic_menu_camera" />

    <!-- ===== SCROLL VIEW CONTENT ===== -->

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Book Details (Scroll View)"
        android:textStyle="bold"
        android:paddingTop="16dp"
        android:paddingBottom="8dp"/>

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="This section demonstrates ScrollView.
```

```
Java Programming book covers OOP concepts, collections, multithreading,
exception handling, and more.
Library Management System helps in managing books, members, issue records, and
returns efficiently.
(Add long description here to see scrolling effect.)
Library Management System helps in managing books, members, issue records, and
returns efficiently."
            android:textSize="16sp"/>
    </LinearLayout>

</ScrollView>
```

- **MainActivity.java**

```
package com.example.practical11;

import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.GridView;
import android.widget.ImageView;
import android.widget.ListView;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    GridView gridView;
    ListView listView;
    ImageView imageView;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // Connect Views
        gridView = findViewById(R.id.gridView);
        listView = findViewById(R.id.listView);
        imageView = findViewById(R.id.imageView);

        // ===== GRID VIEW DATA =====
        String[] dashboard = {
                "Books",
                "Members",
                "Issue",
                "Return"
        };

        ArrayAdapter<String> gridAdapter = new ArrayAdapter<>(
                this,
                android.R.layout.simple_list_item_1,
                dashboard
        );

        gridView.setAdapter(gridAdapter);

        // ===== LIST VIEW DATA =====
        String[] books = {
                "Java Programming",
```

```
                "Android Development",
                "Database Systems",
                "Operating System",
                "Computer Networks"
        };

        ArrayAdapter<String> listAdapter = new ArrayAdapter<>(
                this,
                android.R.layout.simple_list_item_1,
                books
        );

        listView.setAdapter(listAdapter);

        // ===== IMAGE VIEW =====
        imageView.setImageResource(R.drawable.book);
    }
}
```

- **Output:-**